

## UME408 : Machine Design-I

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**Course Objective:** Provide students with the ability to apply design procedure with specific design tools representing empirical, semi-empirical and analytical approaches. Using analytical and computer aided design with real world problems.

### Syllabus

The detailed design of mechanical systems/subsystems based on realistic examples will be followed. Following topics will be covered to understand the design philosophy and to perform the design calculations and failure analysis of various machine elements.

1. Introduction to design of systems and machine elements, design approaches, national and international standards, standard sizes of parts for all the components.
2. Selection of materials.
3. Fits and tolerance, Design for manufacturing
4. Types of loadings, Stresses in machine elements, Stress concentration under static and fluctuating loading, Factor of safety, Modes of failure and theories of failure.
5. Design of Belt Drive and Pulley
6. Design of Coupling
7. Design of Shaft and keys
8. Selection of Rolling contact bearings
9. Design of Spur and Helical Gears
10. Design of threaded fasteners and power screw

**NB:** Prescribed design data book will be allowed in MST and EST. ASTM or equivalent standard will be followed in the design of machine elements.

### Micro Project/ Design Assignment:

In the micro-project, the students will work in groups to redesign angle cutter/ power tool or other mechanical systems. Project activity may include group formation and selection of team leader, communication, dismantling, taking measurements, preparation of questionnaire, feedback from manufacturer/consumer, redesign and reassemble the device/assembly to its original state, computer usage in modeling, drafting and analysis, preparation of final technical report. Use suitable CAD/CAE tools.

Design assignment should include problem formulation, material selection, force analysis, designing of components on the basis of stress analysis and production drawings. Use suitable CAD/CAE tools.

### **Course Learning Objectives (CLO)**

The students will be able to:

1. Select the suitable materials and manufacturing considerations.
2. Calculate stresses involved with static/ fatigue loading
3. Represent machine elements with a free body diagram and solve for unknown reactions
4. Conduct a failure analysis for the design/sizing of mechanical components
5. Design and analyze a real engineering system through projects/assignments

### **Text Books**

1. Shigley, J., Mechanical Engineering Design, McGraw Hill, New York (2014).
2. Bhandari, V. B., Design of Machine Elements, Tata McGraw Hill, New Delhi (2010).

### **Reference Books**

1. Norton, R.L., Machine Design: An Integrated Approach, Pearson Education, New Delhi (2009).
2. Juvinall, R. C. and Marshek, K. M., Fundamental of Machine Component Design, John Wiley & Sons, New York (2011).
3. Sharma, C. S. and Purohil, K., Design of Machine Elements, Prentice Hall, New Delhi (2009).
4. Spotts, M. F. and Shoup, T. E., Design of Machine Elements, Pearson Education, New Delhi (2003).
5. Design Data: Data Book of Engineers by PSG College -Kalaikathir Achchagam – Coimbatore (2012).
6. Mahadevan, K. Balaveera Reddy K., Design Data Handbook for Mechanical Engineers in SI and Metric Units,CBS Publishers & Distributors (2018).

### **Evaluation Scheme**

| <b>S. No.</b> | <b>Evaluation Elements</b> | <b>Weightage (%)</b> |
|---------------|----------------------------|----------------------|
| 1             | MST                        | 30                   |
| 2             | EST                        | 40                   |